

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
image_path = "/content/drogon2.jpg"

original_image = cv2.imread(image_path)

if original_image is None:
    print("Error: Image not found. Check the path.")
else:
    print("Image loaded successfully!")
    print("Image shape:", original_image.shape)
```

```
Image loaded successfully!
Image shape: (260, 520, 3)
```

```
rgb_image = cv2.cvtColor(
    original_image,
    cv2.COLOR_BGR2RGB
)

plt.figure(figsize=(8, 8))
plt.imshow(rgb_image)
plt.title("Original Dragon Image")
plt.axis("off")
plt.show()
```

Original Dragon Image



```
gray_image = cv2.cvtColor(  
    original_image,  
    cv2.COLOR_BGR2GRAY  
)  
  
gray_image_float = np.float32(gray_image)  
  
plt.figure(figsize=(8, 8))  
plt.imshow(gray_image, cmap="gray")  
plt.title("Grayscale Dragon Image")  
plt.axis("off")  
plt.show()
```

Grayscale Dragon Image



```
sobelx = cv2.Sobel(  
    gray_image_float,  
    cv2.CV_64F,  
    1,  
    0,  
    ksize=3  
)  
  
print("Sobel X Min:", sobelx.min())  
print("Sobel X Max:", sobelx.max())
```

```
Sobel X Min: -853.0  
Sobel X Max: 892.0
```

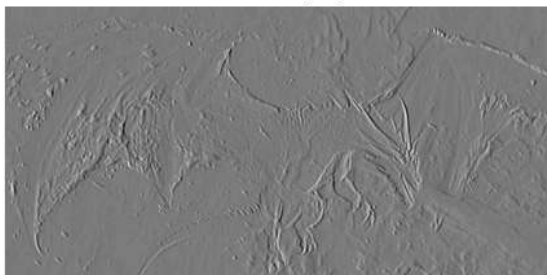
```
sobely = cv2.Sobel(  
    gray_image_float,  
    cv2.CV_64F,
```

```
    0,  
    1,  
    ksize=3  
)  
  
print("Sobel Y Min:", sobely.min())  
print("Sobel Y Max:", sobely.max())
```

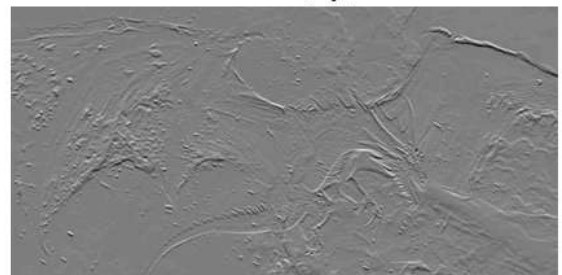
Sobel Y Min: -855.0
Sobel Y Max: 834.0

```
plt.figure(figsize=(12, 5))  
  
plt.subplot(1, 2, 1)  
plt.imshow(  
    sobelx,  
    cmap="gray",  
    vmin=sobelx.min(),  
    vmax=sobelx.max()  
)  
plt.title("Sobel X (Gx)")  
plt.axis("off")  
  
plt.subplot(1, 2, 2)  
plt.imshow(  
    sobely,  
    cmap="gray",  
    vmin=sobely.min(),  
    vmax=sobely.max()  
)  
plt.title("Sobel Y (Gy)")  
plt.axis("off")  
  
plt.show()
```

Sobel X (Gx)



Sobel Y (Gy)



```
gradient_magnitude_sobel = np.sqrt(  
    sobelx**2 + sobely**2  
)
```

```
gradient_magnitude_display_sobel = cv2.normalize(  
    gradient_magnitude_sobel,  
    None,  
    0,  
    255,  
    cv2.NORM_MINMAX  
)  
.astype(np.uint8)  
  
print(  
    "Sobel Gradient Magnitude Min:",  
    gradient_magnitude_sobel.min()  
)  
  
print(  
    "Sobel Gradient Magnitude Max:",  
    gradient_magnitude_sobel.max()  
)
```

Sobel Gradient Magnitude Min: 0.0
Sobel Gradient Magnitude Max: 915.1284062906145

```
gradient_direction_sobel = np.arctan2(  
    sobely,  
    sobelx  
)  
  
gradient_direction_degrees_sobel = np.degrees(  
    gradient_direction_sobel  
)  
  
print(  
    "Sobel Gradient Direction Min:",  
    gradient_direction_degrees_sobel.min()  
)  
  
print(  
    "Sobel Gradient Direction Max:",  
    gradient_direction_degrees_sobel.max()  
)
```

Sobel Gradient Direction Min: -179.8694859216089
Sobel Gradient Direction Max: 180.0

```
gradient_direction_display_sobel = cv2.normalize(  
    (gradient_direction_degrees_sobel + 360) % 360,  
    None,  
    0,  
    255,  
    cv2.NORM_MINMAX  
)  
.astype(np.uint8)
```

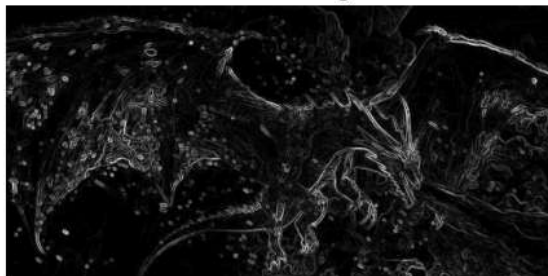
```
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(
    gradient_magnitude_display_sobel,
    cmap="gray"
)
plt.title("Sobel Gradient Magnitude")
plt.axis("off")

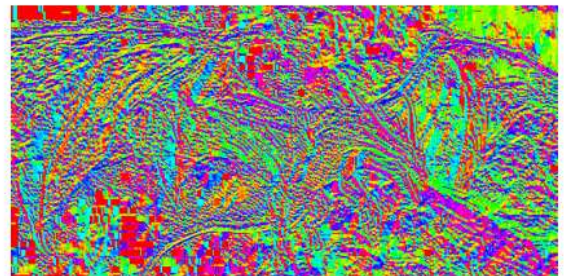
plt.subplot(1, 2, 2)
plt.imshow(
    gradient_direction_display_sobel,
    cmap="hsv"
)
plt.title("Sobel Gradient Direction")
plt.axis("off")

plt.show()
```

Sobel Gradient Magnitude



Sobel Gradient Direction



```
kernelx_prewitt = np.array([
    [-1, 0, 1],
    [-1, 0, 1],
    [-1, 0, 1]
], dtype=np.float32)

kernely_prewitt = np.array([
    [-1, -1, -1],
    [0, 0, 0],
    [1, 1, 1]
], dtype=np.float32)
```

```
Gx_prewitt = cv2.filter2D(
    gray_image_float,
    -1,
    kernelx_prewitt
)
```

```
print("Prewitt Gx Min:", Gx_prewitt.min())  
print("Prewitt Gx Max:", Gx_prewitt.max())
```

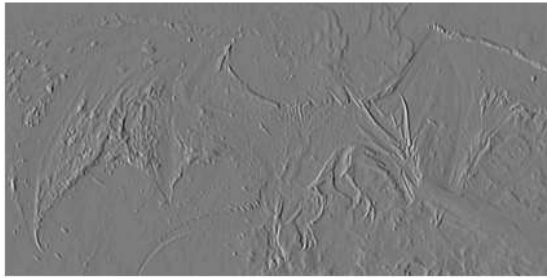
```
Prewitt Gx Min: -635.0  
Prewitt Gx Max: 658.0
```

```
Gy_prewitt = cv2.filter2D(  
    gray_image_float,  
    -1,  
    kernely_prewitt  
)  
  
print("Prewitt Gy Min:", Gy_prewitt.min())  
print("Prewitt Gy Max:", Gy_prewitt.max())
```

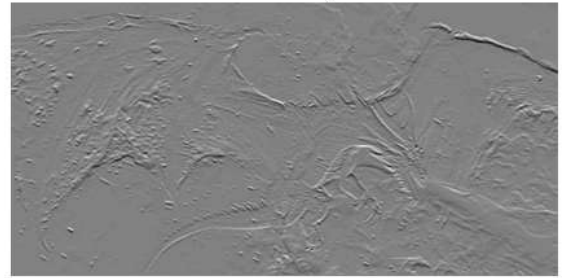
```
Prewitt Gy Min: -642.0  
Prewitt Gy Max: 610.0
```

```
plt.figure(figsize=(12, 5))  
  
plt.subplot(1, 2, 1)  
plt.imshow(  
    Gx_prewitt,  
    cmap="gray",  
    vmin=Gx_prewitt.min(),  
    vmax=Gx_prewitt.max()  
)  
plt.title("Prewitt Gx")  
plt.axis("off")  
  
plt.subplot(1, 2, 2)  
plt.imshow(  
    Gy_prewitt,  
    cmap="gray",  
    vmin=Gy_prewitt.min(),  
    vmax=Gy_prewitt.max()  
)  
plt.title("Prewitt Gy")  
plt.axis("off")  
  
plt.show()
```

Prewitt Gx



Prewitt Gy



```
gradient_magnitude_prewitt = np.sqrt(
    Gx_prewitt**2 + Gy_prewitt**2
)

gradient_magnitude_display_prewitt = cv2.normalize(
    gradient_magnitude_prewitt,
    None,
    0,
    255,
    cv2.NORM_MINMAX
).astype(np.uint8)

print(
    "Prewitt Gradient Magnitude Min:",
    gradient_magnitude_prewitt.min()
)

print(
    "Prewitt Gradient Magnitude Max:",
    gradient_magnitude_prewitt.max()
)
```

```
Prewitt Gradient Magnitude Min: 0.0
Prewitt Gradient Magnitude Max: 660.9145
```

```
gradient_direction_prewitt = np.arctan2(
    Gy_prewitt,
    Gx_prewitt
)

gradient_direction_degrees_prewitt = np.degrees(
    gradient_direction_prewitt
)

print(
    "Prewitt Gradient Direction Min:",
    gradient_direction_degrees_prewitt.min()
)
```



```
print(
    "Prewitt Gradient Direction Max:",
    gradient_direction_degrees_prewitt.max()
)
```

```
Prewitt Gradient Direction Min: -179.85956
Prewitt Gradient Direction Max: 180.0
```

```
gradient_direction_display_prewitt = cv2.normalize(
    (gradient_direction_degrees_prewitt + 360) % 360,
    None,
    0,
    255,
    cv2.NORM_MINMAX
).astype(np.uint8)
```

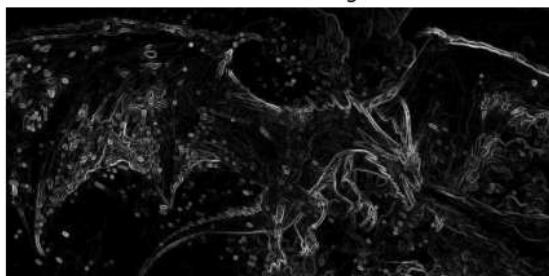
```
plt.figure(figsize=(12, 5))
```

```
plt.subplot(1, 2, 1)
plt.imshow(
    gradient_magnitude_display_prewitt,
    cmap="gray"
)
plt.title("Prewitt Gradient Magnitude")
plt.axis("off")
```

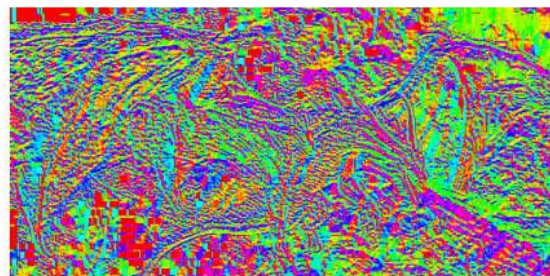
```
plt.subplot(1, 2, 2)
plt.imshow(
    gradient_direction_display_prewitt,
    cmap="hsv"
)
plt.title("Prewitt Gradient Direction")
plt.axis("off")
```

```
plt.show()
```

Prewitt Gradient Magnitude



Prewitt Gradient Direction



```
kernelx_roberts = np.array([
    [1, 0],
```



```
[0, -1]
], dtype=np.float32)

kernelx_roberts = np.array([
    [0, 1],
    [-1, 0]
], dtype=np.float32)
```

```
Gx_roberts = cv2.filter2D(
    gray_image_float,
    -1,
    kernelx_roberts
)

print("Roberts Gx Min:", Gx_roberts.min())
print("Roberts Gx Max:", Gx_roberts.max())
```

```
Roberts Gx Min: -222.0
Roberts Gx Max: 217.0
```

```
Gy_roberts = cv2.filter2D(
    gray_image_float,
    -1,
    kernely_roberts
)

print("Roberts Gy Min:", Gy_roberts.min())
print("Roberts Gy Max:", Gy_roberts.max())
```

```
Roberts Gy Min: -239.0
Roberts Gy Max: 233.0
```

```
plt.figure(figsize=(12, 5))

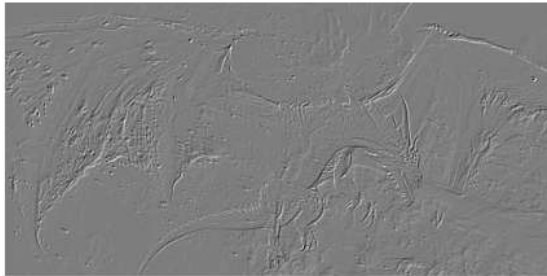
plt.subplot(1, 2, 1)
plt.imshow(
    Gx_roberts,
    cmap="gray",
    vmin=Gx_roberts.min(),
    vmax=Gx_roberts.max()
)
plt.title("Roberts Cross Gx")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(
    Gy_roberts,
    cmap="gray",
    vmin=Gy_roberts.min(),
    vmax=Gy_roberts.max()
```

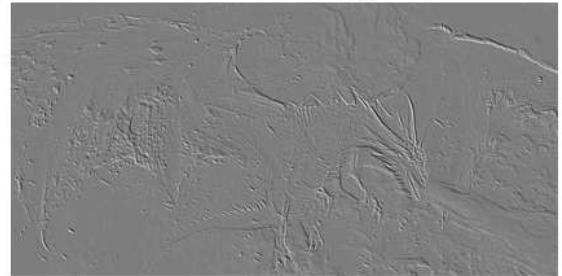
```
)
plt.title("Roberts Cross Gy")
plt.axis("off")

plt.show()
```

Roberts Cross Gx



Roberts Cross Gy



```
gradient_magnitude_roberts = np.sqrt(
    Gx_roberts**2 + Gy_roberts**2
)

gradient_magnitude_display_roberts = cv2.normalize(
    gradient_magnitude_roberts,
    None,
    0,
    255,
    cv2.NORM_MINMAX
).astype(np.uint8)

print(
    "Roberts Gradient Magnitude Min:",
    gradient_magnitude_roberts.min()
)

print(
    "Roberts Gradient Magnitude Max:",
    gradient_magnitude_roberts.max()
)
```

```
Roberts Gradient Magnitude Min: 0.0
Roberts Gradient Magnitude Max: 276.87723
```

```
gradient_direction_roberts = np.arctan2(
    Gy_roberts,
    Gx_roberts
)

gradient_direction_degrees_roberts = np.degrees(
    gradient_direction_roberts
)
```

```
print(
    "Roberts Gradient Direction Min:",
    gradient_direction_degrees_roberts.min()
)

print(
    "Roberts Gradient Direction Max:",
    gradient_direction_degrees_roberts.max()
)
```

```
Roberts Gradient Direction Min: -179.50177
Roberts Gradient Direction Max: 180.0
```

```
gradient_direction_display_roberts = cv2.normalize(
    (gradient_direction_degrees_roberts + 360) % 360,
    None,
    0,
    255,
    cv2.NORM_MINMAX
).astype(np.uint8)

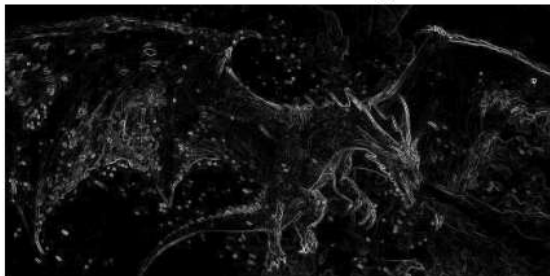
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(
    gradient_magnitude_display_roberts,
    cmap="gray"
)
plt.title("Roberts Cross Gradient Magnitude")
plt.axis("off")

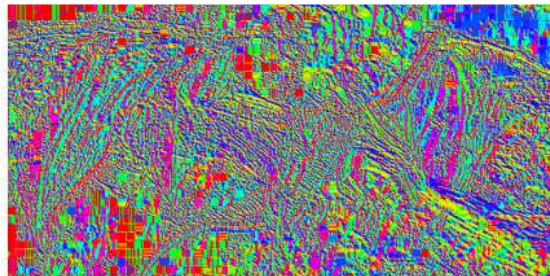
plt.subplot(1, 2, 2)
plt.imshow(
    gradient_direction_display_roberts,
    cmap="hsv"
)
plt.title("Roberts Cross Gradient Direction")
plt.axis("off")

plt.show()
```

Roberts Cross Gradient Magnitude



Roberts Cross Gradient Direction



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